Project Specification: Real-Time Client-Server Messaging System

CSEN 503 — Instructor: Amr Saber

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Version	Date	Summary of changes
V0	9/30/2026	

Project Description

In this project, **teams of four** students will design and implement a real-time messaging application.

At its core, the application will consist of one laptop functioning as the server and each team member's mobile phone acting as a client. To simplify deployment, the server and clients may be connected to the same local Wi-Fi network. The client application can be implemented as a mobile-friendly web interface.

Learning Objectives

This project is designed to provide practical experience that complements your coursework:

- **Network and Link Layer:** Configure endpoint devices within a local network to discover and communicate with one another.
- **Transport Layer:** Build sockets and establish transport-layer connections between devices.
- **Application Layer:** Implement and compare multiple application-layer protocols for message handling.

Deliverables

Each team must schedule a 10-minute appointment with a TA during (teaching) Weeks 11 or 12. The scheduling link will be posted in Week 10 on CMS. During the assessment, your team must present:

- **Live Demo:** A demonstration of team members sending and receiving messages in real time. The demo **must be completed within 5 minutes.** Ensure the system is fully operational before the assessment begins. Timeliness is part of the grading criteria; delays due to technical issues or absent members will result in grade deductions.

  You can use any library available in your selected programming language. For example, you do not have to code a TCP socket from scratch.

- **GitHub Repository:** Provide an online link to your team's GitHub repository. All members must contribute meaningfully. Students with no or anomalously low contributions will receive a zero. This is to ensure fair workload distribution among team members. The application may be implemented in **any programming language(s).**
- **Design Document:** Upload a 3-page PDF (structured in bullet points, not long paragraphs) to the repository. Hand a **physical copy** to your TA before the demo. The document should outline design decisions and include clearly organized sections. Note that the PDF can be the markdown print of your repo's README.
- **Oral Defense:** After the demo, the TA may ask each team member questions about the design and implementation. Each member will have 1 minute to respond. Be prepared to reference code in the repository and content in the PDF. Ensure your code is well-commented, documented, and organized with a clear directory structure. The purpose is to verify your understanding of the design and implementation, and to evaluate the responsible use of LLMs. While LLMs may be used to support coding, you must remain the lead designer, validate outputs, and demonstrate deep understanding of all components. A member can lose up to 100% of their project grade for their inability to coherently answer in the oral defense.

Team Formation

The course project must be completed in **teams of four.** You must submit your teams (complete or incomplete) or individual submissions to this Google form: `https://forms.gle/J3joCjSZBWvRxDmE9`.

After the submission deadline, individuals from incomplete teams will be assigned to form complete teams. Make sure to input your GUC university emails correctly as we will be using the emails you submit to the form to link team members following the submission deadline. It is your responsibility to ensure that your university email inbox is functional (e.g., not full) and can receive emails.

We're allowing cross-tutorial, but not cross-TA: all team members **must be assigned to the same TA.**

After completing the form, you will receive a confirmation email. Make sure you confirm that you have successfully submitted the form by confirming that you have received this confirmation email.

The form submission deadline is Saturday 10th of October 2026 at 06:00 AM Cairo time.

To encourage timeliness and ensure that all teams have been created by October 10, we will have to **enforce a zero project grade for all students who have not made a form submission by the deadline.**

Demo Grading Checklist

Note that while the project requirements and deliverables will remain the same, minor changes to grading might be communicated to you by end of Week 8.

The projects are due by your scheduled appointment with your grading TA.

Network/Link Layer

- All clients successfully connect to the server. ____/1×1

Transport Layer

- Support for simultaneous multiple client connections. ____/1 × 1
- Bi-directional message exchange between all connected clients. ____/1 × 1
- Instant synchronization of messages when clients reconnect from offline mode. For example, a message sent from Phone A should immediately appear in Phone B's chat log when they connect to the server without requiring a manual refresh. ____/1 × 1
- No message loss. ____/1 × 1
- No message duplication. ____/1 × 1

Application Layer

- Correct message encapsulation with no errors. ____/1 × 1
- A responsive, functional, and clean user interface displaying distinct chat logs for different users. ____/1 × 1
- Well-documented, clean, and modular codebase. ____/1 × 1
- Server provides real-time metrics and statistics (e.g., number of active connections, transmission rates) for monitoring and debugging. ____/1 × 1
- Server gracefully handles closed sockets from one client without disrupting other active connections. ____/1 × 1
- Implementation of at least two different application-layer protocols (open choice; one may be custom-designed). ____/1 × 3

  Note that these application layer protocols refer to how you will define the rules, formats, and syntax (. etc.) for exchanging messages between the clients. You may use other application layer protocols beyond these two for other purposes relevant to your implementation.

Additional Functionalities

Beyond the core requirements, each team must implement two additional features of their choice. Examples include:

- User registration and persistent database storage, enabling retrieval of full message history across devices. ____/1 × 3
- Group chat functionality for multi-user communication. ____/1 × 3
- Symmetric end-to-end encryption to secure messages in transit. ____/1 × 3
- Comparative implementation of a web application versus a native mobile application. ____/1 × 3
- Advanced option: Comparison of UDP versus TCP performance. ____/1 × 3
- Advanced option: Peer-to-peer (P2P) communication instead of client–server architecture. ____/1 × 3

Teams are encouraged to propose and implement other creative ideas.

PDF Grading Checklist

The report must outline design decisions that you have implemented in each of the IP stack layers, your additional functionality or functionalities, and your performance metrics and comparisons.

- Clear description of different application layer protocol implementations. ____/1×3
- Clear description of additional functionality. ____/1×3

Performance Evaluation

Each team must report three performance metrics evaluating their implementation for 1 point each. Examples include:

- **Latency:** Average time taken for a message to travel from sender to recipient.
- **Protocol Comparison:** Performance differences between two or more application-layer protocols.
- **Network Conditions:** Impact of device proximity to the Wi-Fi access point (e.g., near vs. far distances).
- **Scalability:** Performance under varying numbers of active connections.
- **Additional functionality comparison:** If implemented, compare UDP vs. TCP, P2P vs. client–server, encrypted vs. unencrypted communication, or web vs. native applications. You need atleast one comparison for your additional functionality.

Presentation

- 3 pages maximum for the PDF report. Single-spaced. Font 11-12. Format is PDF. ____/1×1
- Physical copy and uploaded soft-copy (or README). ____/$1 \times \frac{1}{2}$
- Clearly organized sections and bullet points in the PDF report. ____/$1 \times \frac{1}{2}$
- Well structured code Github repository and documented code. ____/1×3

Contribution

- Commits to Github repo. Deduct 0/50/100% (Similar contribution to other team members/suspiciously less than other team members/none)
- Oral defense. Deduct 0/50/100% (Good coherent answer/weak answer/no coherent answer)

Zero-grade Policy

The following are sure ways to lose all 20% allotted to your project. I am sure you will stay away from them, but they are worth noting:

- No form submission.
- Not working within a team.

- Not working within the team you were assigned to.
- Not having a GitHub repo for your project.
- Not contributing to your team's GitHub repo.
- Unable to show coherent understanding of your product during the oral defense. This might be a symptom of relying on LLMs so much.
- Not doing the project at all.
- No show up: not attending your scheduled project grading slot.
- Disrespect towards your TA during project grading.